Evaluating FAST-LIO2 and Point-LIO for Tree Localization and DBH Estimation in a Brazilian Cerrado Site

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***Abstract*—Accurate pose estimation is essential for mobile robotic forest inventory, as trajectory errors can affect the quality of registered point clouds and dendrometric measurements. This study compares FAST-LIO2 and Point-LIO using identical LiDAR and inertial data collected by a Scout Mini equipped with a Livox MID-360 in a Cerrado area under restoration. Three repetitions of an approximately 10 m $\times$ 7 m route were processed by both methods. Odometry was evaluated through geometric deviation, trajectory closure, path-length deviation, and vertical stability, while the resulting poses were applied to the same tree-mapping pipeline for tree localization and Diameter at Breast Height estimation. FAST-LIO2 and Point-LIO achieved similar planar geometric RMSE values of 0.131 m and 0.132 m, respectively. FAST-LIO2 yielded lower path-length deviation, whereas Point-LIO showed lower vertical variation. In the downstream evaluation, FAST-LIO2 achieved 100% recall, a tree-position RMSE of 34.4 cm, and a DBH RMSE of 5.9 cm, compared with 94.4%, 38.6 cm, and 6.3 cm for Point-LIO. The results show that small differences in LIO outputs can affect the completeness and repeatability of robotic forest inventory.**

***Index Terms*—LiDAR–inertial odometry, mobile forest inventory, tree localization, DBH estimation, FAST-LIO2, Point-LIO.**

I. Introduction

The increase in greenhouse gas emissions intensifies global climate change. As a primary consequence of uncontrolled natural resource exploitation, global warming alters weather and climate patterns, directly affecting human health, food security, and economic stability [1]. Given that forests mitigate this process through carbon sequestration, their conservation and systematic monitoring have become an urgent priority [2], [3]. Dendrometry, the scientific discipline focused on measuring individual trees and forest stands, is therefore essential. By measuring variables such as tree height and Diameter at Breast Height (DBH), allometric equations estimate individual tree biomass and, consequently, carbon [4], [5].

Forest dendrometric data are commonly acquired using Terrestrial Laser Scanning (TLS) or Aerial Laser Scanning (ALS). TLS generates highly accurate point clouds, but requires repeated sensor repositioning, intensive field labor, and prolonged acquisition times. In contrast, ALS efficiently surveys extensive areas; however, canopy occlusion often hampers the accurate measurement of lower stems and trunk geometry [6], [7]. To improve both accuracy and efficiency, recent studies have integrated LiDAR sensors with mobile robotic platforms, allowing them to operate closer to trees while maintaining rapid data acquisition. This approach, known as Mobile Laser Scanning (MLS), enables under-canopy measurements from multiple viewpoints and can substantially reduce acquisition time relative to static TLS [8]. Nevertheless, MLS performance relies on accurate platform localization and is vulnerable to pose-estimation drift. Because Global Navigation Satellite System (GNSS) signals are frequently degraded or unavailable beneath dense forest canopy, LiDAR–inertial odometry techniques are critical for achieving locally accurate and globally consistent point-cloud maps.

FAST-LIO2 and Point-LIO are two state-of-the-art LiDAR–inertial odometry methods based on tightly coupled Kalman-filter formulations [9], [10]. FAST-LIO2 directly registers raw LiDAR measurements to an incrementally maintained map, while Point-LIO performs point-wise state updates and is designed to provide high-frequency estimation under vibration

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and aggressive motion. Nevertheless, their relative performance depends on the sensor, platform dynamics, and environmental structure. Consequently, results from public datasets may not accurately reflect their performance on ground robots operating over irregular terrain and within natural vegetation.

This study compares FAST-LIO2 and Point-LIO using identical LiDAR–inertial recordings from a Scout Mini equipped with a Livox MID-360 in a Cerrado area under ecological restoration. Both methods processed three repetitions of an approximately $10\,\mathrm{m} \times 7\,\mathrm{m}$ route. The pose estimates were evaluated using trajectory metrics and processed through the same tree-mapping pipeline to quantify their impact on tree detection, localization, and DBH estimation.

The main contributions of this study are:

- An experimental comparison of FAST-LIO2 and Point-LIO using a Scout Mini ground robot equipped with a Livox MID-360 in a Cerrado area;
- A trajectory-level evaluation of planar geometric deviation, trajectory closure, path-length deviation, and vertical stability;
- A downstream evaluation of how pose estimates from each method affect tree detection, localization, and DBH estimation when applied to the same tree-mapping pipeline, using field-measured reference trees.

II. Related Work

A. LiDAR–Inertial Odometry

LiDAR–inertial odometry (LIO) combines geometric information from three-dimensional LiDAR measurements with high-frequency motion estimates from an inertial measurement unit. Inertial data support state prediction and motion compensation, while LiDAR measurements correct the accumulated drift by registering the current observations against previous scans or a local map.

FAST-LIO2 is a tightly coupled LIO method based on an iterated Kalman filter [9]. Instead of relying only on extracted geometric features, it directly registers raw LiDAR points against an incrementally maintained map. Its ikd-Tree data structure supports efficient insertion, deletion, downsampling, and nearest-neighbor queries during operation.

Point-LIO also adopts a tightly coupled filtering formulation, but performs state updates for individual LiDAR points [10]. This point-wise strategy avoids waiting for a complete scan and was designed to provide high-bandwidth odometry under vibration, aggressive motion, and possible IMU saturation.

A previous study has already compared these two methods [11] in blueberry crop scenarios. That study evaluated only the translational error at the endpoint of each trajectory, in a structured agricultural setting with a different LiDAR model. Because benchmarks show that LIO methods' relative accuracy and robustness depend on the sensor, motion profile, parameter configuration, and environmental structure [12], an evaluation on the actual robotic platform in the Cerrado environment remains necessary. This work addresses this gap.

B. Mobile LiDAR for Forest Inventory

Forest structure is commonly measured using Terrestrial Laser Scanning (TLS) or Aerial Laser Scanning (ALS). TLS provides dense and accurate observations of stems, but usually requires multiple static acquisitions and considerable field effort. ALS covers larger areas more efficiently, although canopy occlusion and the sensor-to-ground distance can reduce the visibility of lower trunks [6], [7].

Mobile Laser Scanning (MLS) system mounted on ground robots provides measurements below the canopy from multiple viewpoints while the platform moves through the forest. Mobile robotic systems have already been employed for online tree reconstruction and forest inventory [8]. However, the resulting map depends directly on the estimated sensor poses. Pose drift or inconsistent orientation estimates can blur trunk surfaces, duplicate structures, and change the estimated position and geometry of individual trees. Consequently, LIO is not only a localization component but also an important stage in the quality of downstream forest measurements. Therefore, rather than assessing LIO methods by trajectory metrics alone, we also evaluate them through their effect on tree detection and measurement.

C. Tree Detection and DBH Estimation

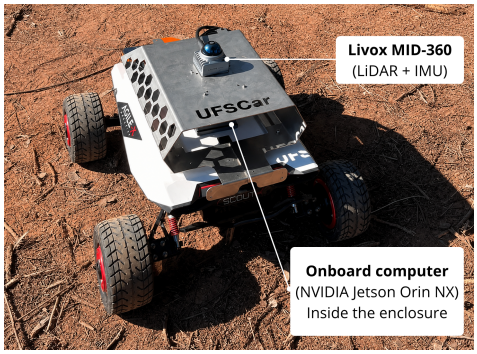
LiDAR-based tree-detection methods commonly combine height filtering, spatial clustering, and robust geometric fitting to identify stem-like structures. Hyyppä et al. [13] process under-canopy UAV point clouds using DBSCAN (Density-Based Spatial Clustering of Applications with Noise) to extract stem candidates and RANSAC (Random Sample Consensus) to fit circular cross-sections, enabling the estimation of tree position and DBH from partially observed trunks. Karjalainen et al. [14] extended this approach toward autonomous under-canopy data acquisition with a UAV, demonstrating its applicability to robotic forest inventory, and Freißmuth et al. [8] proposed an online forest-inventory system based on Mobile Laser Scanning mounted on a robotic platform.

Our tree-perception module builds on these approaches and provides online tree identification and measurement. We treat perception as a black box elsewhere, allowing this work to focus on the impact of the LIO algorithm on its results rather than exploring its specifics.

III. Method

A. Robotic Platform and Study Area

The experimental platform comprised a Scout Mini ground robot as the mobile base, a Livox MID-360 for three-dimensional LiDAR and inertial measurements, and an NVIDIA Jetson Orin NX with 16 GB of memory for onboard data acquisition. The LiDAR was mounted on the upper enclosure, while the embedded computer was installed inside a protected compartment. The Jetson executed the ROS 2 Humble software stack, communicated with the mobile base through a USB-CAN interface, and recorded the experimental data (Fig. 1(a)).



(a)



(b)

Fig. 1. Experimental platform and study area. In (a) Scout Mini equipped with a Livox MID-360 and an onboard NVIDIA Jetson Orin NX and (b) Cerrado study area in the approximate rectangular trajectory used during data acquisition.

Data acquisition was conducted in a Cerrado area under restoration, with tree stems of different sizes and irregular terrain. The experiments were performed along a marked rectangular path, with the robot starting from the same reference region in each run (Fig. 1(b)). A representative experimental run is shown in the supplementary video.[1]

An approximately $10\,\mathrm{m} \times 7\,\mathrm{m}$ rectangular path was physically marked on the ground using a string, resulting in a nominal perimeter of $34\,\mathrm{m}$. The Scout Mini was manually teleoperated along the marked boundary until it returned to the starting point. The dashed line in Fig. 1(b) was added to indicate the approximate path because the physical string was not clearly visible from the camera viewpoint.

The experiment was repeated three times under the same operating conditions. In each run, the robot started from approximately the same reference region and completed one lap. Due to the manual operation, rounded turns, terrain irregularities, and the dimensions of the mobile platform, the executed trajectories did not perfectly reproduce a rectangle. Therefore, the marked path was considered an approximate geometric reference rather than a centimeter-accurate ground-truth trajectory.

[1]https://youtu.be/9koUECC1Gzs

B. Data Acquisition and Pose Estimation

During each experimental run, all sensor and navigation data were recorded in a single ROS 2 bag. The dataset included the Livox MID-360 point cloud, inertial measurements, wheel odometry, and commanded robot velocities. The timestamps stored in the message headers were used for trajectory extraction and synchronization.

Each of the three Cerrado bags was processed independently using FAST-LIO2 and Point-LIO. Both algorithms received the same LiDAR and inertial measurements and used identical LiDAR–IMU extrinsic parameters. The translation was set to $[-0.011, -0.02329, 0.04412]$ m, the relative rotation was the identity matrix, and online extrinsic estimation was disabled. Point-LIO preprocessing was locally adapted to read the Livox MID-360 `PointCloud2` field structure, including the per-point timestamp, without requiring an external conversion of the recorded bag.

The FAST-LIO2 trajectory was extracted from the `/Odometry` topic, with `camera_init` as the reference frame and `body` as the child frame. The Point-LIO trajectory was obtained from `/aft_mapped_to_init`, using `camera_init` and `aft_mapped` as its reference and child frames, respectively. Both outputs use the `nav_msgs/msg/Odometry` message type and were treated as the pose estimates produced by the corresponding LIO method.

The wheel-odometry trajectory used the `odom` and `base_link` frames. It was included only as an auxiliary proprioceptive baseline for trajectory visualization and analysis. Wheel odometry was neither provided as input to FAST-LIO2 or Point-LIO nor used to generate the point-cloud maps employed for tree localization and DBH estimation.

For quantitative comparison, the LIO trajectories were expressed in a common planar relative representation initialized at the first estimated pose. Only planar translation and yaw rotation were allowed during alignment; no scale correction or local trajectory deformation was applied. The three runs, therefore, produced six LiDAR–inertial trajectories, together with three wheel-odometry baseline trajectories.

C. Tree Mapping and DBH Estimation

The tree-mapping stage was applied online, consuming the pose estimate from the LiDAR–inertial odometry method. The same mapping parameters were adopted for FAST-LIO2 and Point-LIO so that the downstream comparison reflected differences in the reconstructed maps rather than in the tree-mapping algorithm.

The proposed method, built on [13], identifies vertical structures that exhibit the geometric characteristics of tree stems. First, the point cloud is transformed into a fixed reference frame and filtered to retain the height range expected for trunks, with points near the robot being removed. The remaining cloud is divided into horizontal slices, where DBSCAN groups neighboring points and rejects sparse measurements as

noise. For each cluster, RANSAC fits a circular cross-section. Instead of performing a full three-dimensional cylinder fit, the method represents each stem with circular fits in successive horizontal slices. A candidate is classified as a tree only when the estimated circles remain geometrically consistent along the vertical direction.

After classification, tree detections are tracked and fused over time. A tree is incorporated into the final map only after repeated observations with consistent position and estimated diameter. The diameter estimate obtained from the accepted circular-stem evidence is used as the DBH estimate.

IV. Experimental Evaluation

The experimental evaluation investigates how the choice of LiDAR–inertial odometry method affects both robot localization and downstream forest-inventory measurements. FAST-LIO2 and Point-LIO were applied under identical experimental conditions. The comparison focuses on two aspects: (i) trajectory consistency and preservation of the traversed-path geometry; and (ii) the impact of pose-estimation differences on the tree detection, localization, and DBH estimation.

A. Evaluation Protocol and Ground-Truth Measurements

Six trees, denoted GT1–GT6, were selected as reference trees measured in the field. Their planar positions were manually measured with a measuring tape relative to a local origin established in the experimental area. Both estimated and reference positions were expressed in this common local coordinate system.

For each reference tree, the circumference at breast height (CBH) was measured at approximately $1.30\,\mathrm{m}$ above the ground. The corresponding reference DBH was calculated as

$$d_i^{\mathrm{ref}} = \frac{C_i^{\mathrm{ref}}}{\pi}, \tag{1}$$

where C_i^{ref} is the measured circumference and d_i^{ref} is the reference DBH of tree i.

Estimated trees were paired with field reference trees according to their planar distance. A detection was considered a match if it was located within a radius of $2.5\,\mathrm{m}$ of the corresponding field position.

For each matched tree i, the planar position error was calculated as

$$e_{p,i} = \sqrt{\left(x_i - x_i^{\mathrm{ref}}\right)^2 + \left(y_i - y_i^{\mathrm{ref}}\right)^2}, \tag{2}$$

where (x_i, y_i) is the planar position of matched tree i estimated by the tree-mapping pipeline, and $(x_i^{\mathrm{ref}}, y_i^{\mathrm{ref}})$ is its field-measured reference position, both expressed in the local coordinate system of the experimental area. The tree-position RMSE was then obtained from

$$\mathrm{RMSE}_p = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_{p,i}^2}, \tag{3}$$

where N is the number of matched trees. The DBH RMSE was calculated as

TABLE I
Odometry results across the three Cerrado runs. Values are reported as mean ± sample standard deviation.

Metric	FAST-LIO2	Point-LIO
Geometric deviation RMSE (m)	0.131 ± 0.027	0.132 ± 0.026
Trajectory closure error (m)	0.118 ± 0.048	0.131 ± 0.015
Path-length deviation (m)	2.778 ± 1.014	5.062 ± 1.994
Final vertical drift (m)	0.020 ± 0.005	0.011 ± 0.009
Vertical peak-to-peak (m)	0.603 ± 0.150	0.194 ± 0.016

$$\mathrm{RMSE}_{\mathrm{DBH}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\widehat{d}_i - d_i^{\mathrm{ref}}\right)^2}, \tag{4}$$

where $\widehat{d}_i$ denotes the DBH estimated by the tree-mapping pipeline. Detection performance was summarized by the number of confirmed tree instances and recall relative to the six field-reference trees.

B. Odometry Performance

Figure 2 compares the trajectories estimated by FAST-LIO2 and Point-LIO in the three Cerrado runs. Wheel odometry is included only as an auxiliary baseline, and the dashed rectangle indicates the approximate nominal $10\,\mathrm{m} \times 7\,\mathrm{m}$ path. All trajectories are expressed relative to their initial pose, without scale correction.

The trajectory-closure error was defined as the planar distance between the initial and final estimated positions, $e_c = \sqrt{(x_N - x_1)^2 + (y_N - y_1)^2}$. The path-length deviation was computed as $|L_{\mathrm{est}} - 34|$ m, where L_{est} is the estimated two-dimensional trajectory length and 34 m is the nominal perimeter. Final vertical drift was defined as $|z_N - z_1|$, whereas vertical peak-to-peak variation was computed as $\max(z) - \min(z)$. Because the terrain was not perfectly planar, the latter is interpreted as vertical variation rather than an absolute localization error.

Both LIO methods produced similar planar trajectories and remained substantially closer to the nominal rectangular geometry than wheel odometry. Their geometric-deviation values were also consistent across the three runs, indicating no clear advantage in planar trajectory accuracy for either method.

Table I summarizes the main odometry metrics for the three runs. FAST-LIO2 produced path-length estimates consistently closer to the nominal $34\,\mathrm{m}$ perimeter, whereas Point-LIO exhibited lower vertical variation. Trajectory-closure errors varied between runs but remained of the same order of magnitude for both methods. Overall, these differences were small, motivating the subsequent analysis of their impact on tree localization and DBH estimation.

C. Tree Localization and DBH Estimation

Each execution of the tree-mapping pipeline generated a map containing the detected tree locations and their corresponding DBH estimates. Figure 3 presents an example generated from the FAST-LIO2 trajectory. The registered point cloud is overlaid with the cylindrical models produced by

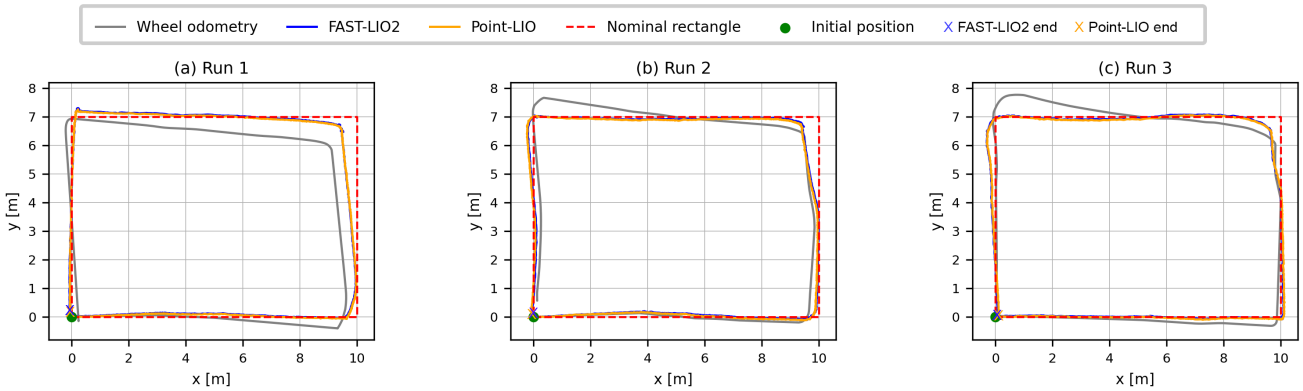


Fig. 2. Normalized trajectories obtained in the three Cerrado runs. FAST-LIO2 and Point-LIO were evaluated using identical experimental conditions. The dashed rectangle denotes the approximate nominal path, while wheel odometry is shown only as a comparison.

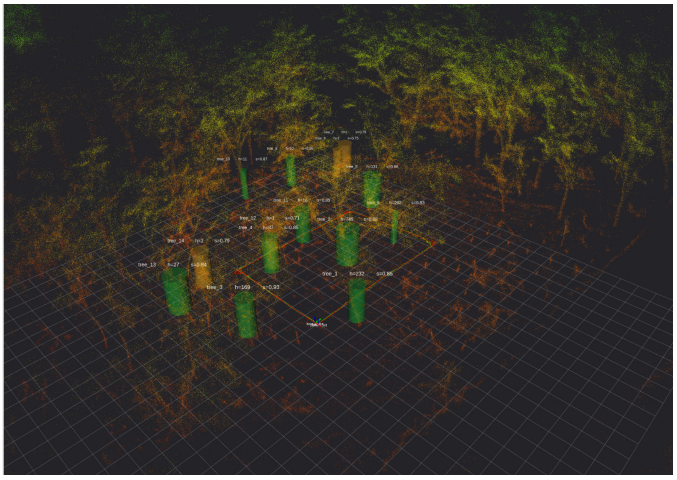


Fig. 3. Overlap view of the consolidated point cloud and the cylindrical model generated for detected trees. Green cylinders represent confirmed trees, whereas yellow cylinders indicate candidates. The estimated FAST-LIO2 trajectory is also shown. The score assigned to each candidate is computed from the circle-fitting results by combining the number of inliers with a penalty based on the mean fitting error. (H) denotes the number of hits, i.e., the number of times the corresponding tree was observed.

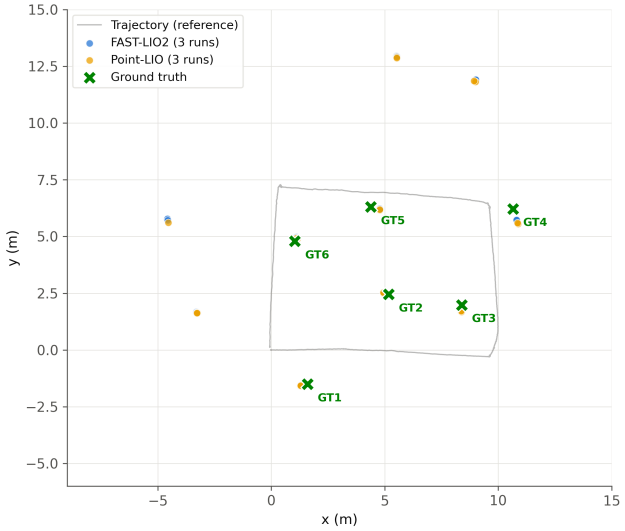


Fig. 4. Tree positions estimated over the three experimental runs using FAST-LIO2 and Point-LIO odometry. Green crosses denote the field-reference positions. The concentration of repeated estimates around each reference tree illustrates the spatial repeatability of the tree-mapping pipeline.

the tree-mapping node. Confirmed trees are shown in green, whereas candidates that did not satisfy the temporal confirmation criteria are shown in yellow.

Figure 4 compares the field-reference tree positions, represented by green crosses, with the positions estimated in the three runs using FAST-LIO2 and Point-LIO. For both methods, repeated detections associated with the same reference tree remain concentrated in similar regions, indicating good spatial repeatability. Nevertheless, FAST-LIO2 exhibits lower spatial dispersion and estimates that are generally closer to the reference positions. Additional trees detected by the algorithm were not included in the field-reference map and, therefore, excluded from the quantitative evaluation. These detections correspond to the orange and blue markers without a matching green cross in Fig. 4.

Table II summarizes the tree-detection, localization, and DBH-estimation results for the six ground-truth reference trees (GT1–GT6). FAST-LIO2 detected all six reference trees in every run, achieving 100% recall with no run-to-run variation. Point-LIO also achieved 100% recall in the first two runs but detected only five of the six trees in the third run (83.3%), missing GT5.

TABLE II
AGGREGATED PERFORMANCE ACROSS THE THREE EXPERIMENTAL RUNS.

Metric	FAST-LIO2	Point-LIO
Confirmed tree instances	18/18	17/18
Overall recall (%)	100.0	94.4
Mean position RMSE (cm)	34.4	38.6
Mean DBH RMSE (cm)	5.9	6.3
Mean odometry output rate (Hz)	7.6	3.2

The lower recall achieved by Point-LIO was associated with reduced point support. For trees detected by both methods, Point-LIO accumulated three to five times fewer stem observations than FAST-LIO2, consistent with its lower odometry output rate (Table II). Because the tree-mapping pipeline relies on the published odometry from the active LIO to transform points into world coordinates and requires a minimum number of repeated detections before confirming a tree, a lower output frequency reduces opportunities to accumulate evidence over the same traversal. This effect was most evident for GT5, the smallest and least-observed reference tree, which accumulated only 4, 3, and 0 detections with Point-LIO across the three runs, compared to 16, 26, and 27 with FAST-LIO2.

FAST-LIO2 also produced lower and more consistent position errors. Mean DBH RMSE was similar for both methods ($5.9\,\mathrm{cm}$ for FAST-LIO2 vs. $6.3\ \mathrm{cm}$ for Point-LIO), although FAST-LIO2 exhibited higher repeatability, with a variation of only $0.6\,\mathrm{cm}$ across runs (5.7–$6.3\,\mathrm{cm}$), compared with $1.7\ \mathrm{cm}$ for Point-LIO (5.4–$7.1\,\mathrm{cm}$).

Overall, FAST-LIO2 provided the more reliable input to the tree-mapping pipeline in the evaluated experiments, achieving complete recall, lower position and DBH RMSE, and smaller run-to-run variation. However, the downstream differences cannot be attributed solely to pose-estimation accuracy, since both methods achieved similar planar trajectory errors and differed in the rate at which odometry updates were made available to the mapping pipeline. Therefore, the observed performance reflects the combined effects of pose quality and temporal availability of the odometry estimates.

V. Conclusion and future work

This work compared FAST-LIO2 and Point-LIO using identical LiDAR and inertial recordings acquired by a ground robot in a Cerrado area under restoration. The methods produced nearly identical planar geometric RMSE, while FAST-LIO2 achieved smaller path-length and closure errors, and Point-LIO exhibited lower vertical variation. Therefore, neither method was superior in every odometry metric.

For the downstream forest-inventory task, FAST-LIO2 provided the more reliable input. It confirmed all 18 expected tree instances, achieved 100% recall, and obtained lower mean position and DBH RMSE values. Point-LIO confirmed 17 instances and achieved comparable DBH accuracy for detected trees, but its lower output rate reduced the observations available for temporal confirmation. These results demonstrate that trajectory metrics alone do not fully characterize the suitability of an LIO method for robotic forest inventory.

The evaluation was limited to three runs, six reference trees, one study area, and an approximate trajectory reference. Also, the LIO algorithms are environment-dependent, and the results may not generalize to other scenarios. Future work will include larger forest areas, autonomous trajectories, survey-grade reference measurements, and computational comparisons under identical replay conditions. The pose-update frequency received by the tree-mapping pipeline will also be normalized to separate the effects of pose accuracy, temporal resolution, and processing cost.

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